

BEREKET, Turkmenistan

WMO: 386470

Lat: 39.250N

Lon: 55.517E

Elev: 31

StdP: 100.95

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.7	-5.9	-16.4	0.9	-0.2	-13.6	1.2	1.5	13.5	3.3	12.2	0.5	6.4	90	0.511

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	41.1	20.2	39.7	20.1	38.3	20.1	24.5	34.7	23.8	33.9	23.1	33.1	4.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	16.2	30.5	20.4	15.1	30.5	19.3	14.1	30.7	74.5	34.6	71.5	34.0	68.8	33.1	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.1	10.4	9.2	-10.3	44.2	5.4	1.6	-14.3	45.3	-17.4	46.3	-20.5	47.2	-24.5	48.4		
			-11.2	25.8	5.3	1.1	-15.0	26.6	-18.1	27.3	-21.1	27.9	-24.9	28.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.6	3.2	4.7	10.9	17.5	24.5	29.7	32.1	31.3	25.2	17.4	9.5	4.5	(6)
(7)		DBStd	11.16	4.57	5.21	4.36	4.61	3.40	2.94	2.40	2.61	3.42	4.42	4.76	4.10	(7)
(8)		HDD10.0	651	212	154	39	3	0	0	0	0	4	63	175		(8)
(9)		HDD18.3	1922	468	382	234	69	3	0	0	2	70	267	429		(9)
(10)		CDD10.0	3433	2	6	68	228	449	589	687	659	457	234	48	5	(10)
(11)		CDD18.3	1661	0	0	3	44	193	339	428	401	209	42	2	0	(11)
(12)		CDH23.3	22611	0	0	64	450	2066	4775	6576	6025	2279	360	17	0	(12)
(13)		CDH26.7	13173	0	0	18	161	954	2838	4219	3803	1074	101	4	0	(13)
(14)	Wind	WSAvg	3.7	4.1	4.3	3.9	3.8	3.2	3.4	3.4	3.3	3.8	3.6	3.8	4.2	(14)
(15)	Precipitation	PrecAvg	171	21	21	27	22	14	2	5	3	2	15	17	25	(15)
(16)		PrecMax	311	47	66	75	61	74	12	28	36	16	81	37	73	(16)
(17)		PrecMin	77	3	0	2	0	0	0	0	0	0	0	0	1	(17)
(18)		PrecStd	53	11	17	17	16	15	4	10	7	3	18	12	21	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.9	20.3	29.2	34.2	38.3	41.7	43.2	38.2	32.4	25.5	19.7		(19)
(20)		0.4%	MCWB	9.8	11.0	14.6	17.3	18.3	20.0	20.9	20.2	18.1	17.5	14.6	10.6	(20)
(21)		2%	DB	13.6	16.1	24.7	30.7	35.5	39.9	41.3	41.0	35.6	29.3	20.9	15.1	(21)
(22)		2%	MCWB	8.2	9.6	13.2	16.6	18.1	19.9	20.7	19.7	18.4	16.3	13.1	9.6	(22)
(23)		5%	DB	10.9	13.5	21.1	28.1	33.5	38.3	39.8	39.2	33.9	27.1	18.5	12.4	(23)
(24)		5%	MCWB	7.2	8.4	12.2	15.6	17.8	19.6	20.6	19.9	18.2	15.6	11.8	8.3	(24)
(25)		10%	DB	8.9	11.5	18.2	25.7	31.6	36.7	38.4	37.8	32.1	25.0	16.4	10.4	(25)
(26)		10%	MCWB	6.0	7.4	11.0	14.7	17.3	19.5	20.8	19.9	17.7	14.7	11.1	7.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.6	12.0	15.9	19.1	20.9	24.7	25.6	25.5	23.3	19.9	15.5	11.6	(27)
(28)		0.4%	MCDWB	15.5	17.9	25.7	30.1	32.0	36.7	36.2	35.3	32.0	28.2	22.7	17.1	(28)
(29)		2%	WB	8.7	10.2	14.0	17.6	19.9	22.9	24.6	24.5	21.9	18.4	14.0	9.9	(29)
(30)		2%	MCDWB	12.8	15.0	21.8	28.3	30.6	34.5	34.7	34.6	30.9	26.3	19.6	14.2	(30)
(31)		5%	WB	7.4	8.8	12.7	16.6	19.2	21.9	23.9	23.7	20.7	17.2	12.7	8.7	(31)
(32)		5%	MCDWB	10.5	12.7	20.1	26.2	29.5	33.8	34.2	33.7	30.0	24.3	17.5	11.8	(32)
(33)		10%	WB	6.2	7.7	11.4	15.5	18.6	21.0	23.3	22.9	19.5	15.9	11.6	7.6	(33)
(34)		10%	MCDWB	8.8	11.0	17.4	23.9	28.6	32.2	33.4	32.7	29.5	22.9	15.6	10.3	(34)
(35)	Mean Daily Temperature Range	MDBR	7.5	8.4	10.3	11.0	11.6	11.9	11.1	12.0	12.2	11.7	9.0	7.5	(35)	
(36)		5% DB	MCDBR	11.7	13.1	15.7	15.4	14.3	14.1	13.3	14.1	14.2	14.3	12.4	11.8	(36)
(37)		5% DB	MCWBR	7.2	7.5	7.1	5.9	4.8	5.2	4.8	5.4	6.1	6.5	6.5	7.4	(37)
(38)		5% WB	MCDBR	10.7	11.7	13.8	13.8	11.9	11.7	10.6	10.2	11.2	11.7	10.7	10.2	(38)
(39)		5% WB	MCWBR	6.9	7.0	6.7	5.7	4.5	4.9	4.4	5.0	6.1	6.0	6.1	6.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.410	0.461	0.509	0.579	0.559	0.560	0.584	0.539	0.518	0.501	0.443	0.416	(40)	
(41)		taud	2.027	1.897	1.816	1.710	1.785	1.793	1.753	1.833	1.849	1.870	1.985	2.011	(41)	
(42)		Ebn at Noon	711	727	739	717	744	743	720	741	722	676	665	664	(42)	
(43)		Edh at Noon	124	162	194	230	217	216	223	201	186	164	128	118	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.15	2.98	4.18	5.30	6.71	7.66	7.32	6.82	5.60	3.89	2.62	1.92	(44)	
(45)		RadStd	0.20	0.33	0.28	0.54	0.32	0.22	0.29	0.33	0.20	0.21	0.19	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page